

Ian Hussey

Institute of Psychology
University of Bern
Fabrikstrasse 8, Bern 3012
Switzerland

Email: ian.hussey@unibe.ch
Google Scholar: [Sg6JZb0AAAAJ](https://scholar.google.com/citations?user=Sg6JZb0AAAAJ)
ORCID: [0000-0001-8906-7559](https://orcid.org/0000-0001-8906-7559)
GitHub: github.com/ianhussey

Employment and appointments

- 2025 – present Senior Lecturer (Dozent I)
Psychology of Digitalisation
Institute of Psychology
University of Bern
- 2024 Visiting Research Fellow
Oxford Internet Institute
University of Oxford
- 2023 – 2025 Postdoctoral Researcher
Psychology of Digitalisation
Institute of Psychology
University of Bern
- 2022 – 2023 Postdoctoral Researcher
Department of Psychology of Human-Technology Interaction
Faculty of Psychology
Ruhr University Bochum
- 2015 – 2020 Postdoctoral Researcher
Learning and Implicit Processes Lab
Department of Experimental, Clinical and Health Psychology
Ghent University

Education

- 2025 Habilitation (PD)
Thesis title: 'Meta-science and the credibility revolution'
University of Bern
- 2010 – 2015 Ph.D.
Thesis title: 'Developing the Implicit Relational Assessment Procedure (IRAP) to Assess Evaluations of Death Within Suicidality'
Maynooth University
- 2013 Professional Certificate in Teaching and Learning
Maynooth University
- 2007 – 2010 B.A. (Hons.) Psychology
First Class Honours (1.1)
Maynooth University

Research interests

Meta-science, meta-analysis, trustworthiness assessment; psychological measurement practices; development and validation of statistical methods; simulation studies, AI for psychology research.

Preprints and under-review manuscripts

2026

- Cummins, J., Clarke, B., **Hussey, I.**, & Elson, M. (2026, under review). RegCheck: A tool for automating comparisons between study registrations and papers.
<https://arxiv.org/pdf/2601.13330>
- Cummins, J., Drysdale, H., Elson, M., **Hussey, I.**, Goldacre, B., & DeVito, N. (2026). Automating clinical trial outcome identification and misreporting detection using RegCheck. *medRxiv*. <https://doi.org/10.64898/2026.07.30.26358885>
- Freeman, Z., & **Hussey, I.** (2026). Item-dropping to improve in-sample Cronbach's alpha can harm true reliability: A Monte Carlo simulation. *PsyArXiv*.
<https://osf.io/preprints/psyarxiv/7jc3t>
- Heal, C., Bero, L., Antoniou, G. A., Au, N., Aviram, A., Berghella, V., Bordewijk, E. M., Bramley, P., Brown, N. J. L., Clarke, M., Fiala, L., Grohmann, S., Gurrin, L. C., Hayden, J. A., Hunter, K. E., **Hussey, I.**, Kahan, B. C., Lensen, S., Lundh, A., O'Connell, N. E., Parker, L., Lam, E., Meyerowitz-Katz, G., Naudet, F., Redman, B. K., Sheldrick, K., Sydenham, E., van Wely, M., Wang, R., Wjst, M., Kirkham, J., & Wilkinson, J. (2026). Protocol for the development of a tool (INSPECT-IPD) to identify problematic randomised controlled trials when individual participant data are available. *medRxiv*.
<https://doi.org/10.64898/2026.02.06.26345217>
- Hussey, I.** (2026, under review: *Journal of Replication Research*). No evidence of differences between hierarchical versus distinction relations in self-based ACT exercises: A critical reanalysis of Foody et al. (2013) and Foody et al. 2015). *PsyArXiv*.
https://doi.org/10.31234/osf.io/ab7de_v1
- Hussey, I.** (2026). Very little data tampering is needed to produce statistically significant results: A Monte Carlo simulation. *PsyArXiv*. <https://osf.io/preprints/psyarxiv/7jc3t>
- Hussey, I.**, & Elson, M. (2026, under review: *Journal of Contextual Behavioral Science*). The RFT Perspective Taking Protocol has very poor reliability and convergent validity with other measures of perspective taking, empathy, and theory of mind: A jingle assessment. doi: https://doi.org/10.31234/osf.io/jb6c4_v1
- Hussey, I.**, Jung, L., Lakena, W., & Cummins, J. (2026). The BRIMMER test: The bounds of the Standard Deviation under constraints. *PsyArXiv*.
<https://osf.io/preprints/psyarxiv/4v7pe>
- Külpmann, A. I., Ries, J. P., **Hussey, I.**, & Elson, M. (2026,; *Advances in Methods and Practices in Psychological Science*). Methodological Flexibility in the Iowa Gambling Task Undermines Interpretability: A Meta-method review. *PsyArXiv*.
https://doi.org/10.31234/osf.io/4g3vr_v1
- Norwood, S. F., **Hussey, I.**, Grüning, D. J., Elson, M., & Przybylski, A. K. (2026, under review: *Nature Human Behaviour*). Digital behaviourism: A functional approach to behaviour in digital environments. https://doi.org/10.31234/osf.io/ztev4_v2

2025

- Alsalti, T., **Hussey, I.**, Elson, M., Arslan, R.C. (2025, under review: *Behavior Research Methods*). An R/brms tutorial on using Regularised Prediction and Poststratification for test norming. https://doi.org/10.31234/osf.io/fcm3n_v3
- Harrer, M., Miguel, C., **Hussey, I.**, Cristea, I. A., van Ballegooijen, W., Basic, D., Wang, Y., Pfund, R. A., Quero, S., von Spreckelsen, P., Schnurr, P. P., van Straten, A., Furukawa, T. A., Papola, D., Cuijpers, P. (2025) Implausible Effects of Psychological Interventions: Meta-Epidemiological Study and Development of a Simple Flagging Tool. *medRxiv*. <https://doi.org/10.1101/2025.11.12.25340062>
- Hussey, I.** (2025). Trustworthiness assessment of Gloster et al. (2020) ‘Treating treatment non-responders: A meta-analysis of randomized controlled psychotherapy trials’. https://doi.org/10.31234/osf.io/rbydj_v2
- Norwood, S. F., Elson, M., & **Hussey, I.** (2025, under review: *PCI Registered Reports*). Psychology: The science of behavior? [Registered Report stage 1 submission] <https://osf.io/ykw4s/files/cxw8k>

2024 and prior

- Hussey, I.** (2023). The Implicit Relational Assessment Procedure’s trial-types are not independent. <https://osf.io/preprints/psyvarxiv/4gsxq>
- Hussey, I.** (2023). Reply to Barnes-Holmes & Harte (2022) ‘The IRAP as a Measure of Implicit Cognition: A Case of Frankenstein’s Monster.’ <https://psyvarxiv.com/qmg6s>
- Hussey, I.** (2022). The IRAP is not suitable for individual use due to very wide confidence intervals around *D* scores. <https://psyvarxiv.com/w2ygr>
- Hussey, I.**, & Drake, C. E. (2022). The IRAP is not very sensitive to the attitudes and learning histories it seeks to assess. <https://psyvarxiv.com/sp6jx>
- Hussey, I.**, & Drake, C. E. (2022). The Implicit Relational Assessment Procedure demonstrates poor internal consistency and test-retest reliability: A meta-analysis. <https://psyvarxiv.com/ge3k7>
- Hussey, I.**, & Cummins, J. (2022). Evidence against effects on the Affect Misattribution Procedure being unaware: AMP effects involve construct-irrelevant individual differences. <https://psyvarxiv.com/8k94v>
- Greenwald, A. G., Dasgupta, B., Dovidio, J., Hofmann, W., Kang, J., Moss-Racusin, C., Steele, C., Teachman, B. A., Brendl, M., Cai, H., Charlesworth, T., Cvenček, D., Dovidio, J., Hahn, A., Hehman, E., Hofmann, W., Hughes, S., **Hussey, I.**, Jordan, C., Jost, J., Kirby, T., Kurdi, B., Lai, C., Lang, J., Lindgren, K. P., Maison, D., Ostafin, B., Rae, J., Ratliff, K., Smith, C. T., Spruyt, A., Wiers, R. (2020, elements published as Greenwald et al., 2022). The Implicit Association Test at age 20: What is known and what is not known about implicit bias. <https://psyvarxiv.com/bf97c> [authorship order determined alphabetically]
- Hussey, I.** (2020). General claims require generalized effects: A reply to Ruiz et al.’s (2020) ‘A systematic and critical response to Pendrous et al. (2020) replication study’. <https://psyvarxiv.com/83z2y>
- Murthy, V. E., **Hussey, I.**, & McHugh, L. (2020). A single-session Acceptance and Commitment Therapy intervention targeting shame in people experiencing homelessness: A randomised multiple baseline design. <https://psyvarxiv.com/3hpug>
- Hussey, I.** & De Houwer, J. (2018). Completing a Race IAT increases implicit racial bias. <https://psyvarxiv.com/vxsi7>

Peer-reviewed publications & book chapters

2026

- Alsalti, T., **Hussey, I.**, Elson, M., Krause, R., & Pohl, S. (2026). A Methodological Evaluation of Meta-Analyses in tDCS-Motor Learning Research. *European Journal of Neuroscience* 63(9), e70533 <https://doi.org/10.1111/ejn.70533>
- Alsalti, T., **Hussey, I.**, Elson, M., & Arslan, R.C. (2026). A Case Study on Norming an Intelligence Test Using Regularised Prediction and Poststratification. *Collabra*, 2(1), 155850. <https://doi.org/10.1525/collabra.155850>
- Borgert, N., **Hussey, I.**, Jansen, L., & Elson, M. (2026) Do I Value My Private Data? Exploring Model-Driven Predictions on the Willingness to Use Smart Home Devices. *Media Psychology*. <https://doi.org/10.1080/15213269.2025.2463392>
- Cummins, J. & **Hussey, I.** (2026). The individual-level precision of implicit measures. *Behavior Research Methods*, 58, 21. <https://doi.org/10.3758/s13428-025-02873-2>
- Elson, M., **Hussey, I.**, Clarke, B., Norwood, S. F., Grinschgl, S., Arslan, R. C., Wilkinson, J., Page, M. J., Cristea, I. A., Cuijpers, P., Vazire, S., DeVito, N. J., & Cummins, J. (2026, *in press*). Against Anonymising Meta-Scientific Data. *Advances in Methods and Practices in Psychological Science*. https://doi.org/10.31234/osf.io/6eyjf_v2
- Plöderl, M., **Hussey, I.**, & Ostacher, M. (2026). Trustworthiness issues in Yari-Renani and Zari's (2025) study “Advancing depression treatment: a multi-center randomized controlled trial of dynamic interpersonal therapy versus CBT and pharmacotherapy on symptoms, sleep, and cognition”. *Psychiatry Research*. <https://doi.org/10.1016/j.psychres.2026.116985>
- Wilkinson, J., Heal, C., Fleming, E., ... **Hussey, I.**, ... & Kirkham, J. J. (2026, *in press*). INSPECT-SR: a tool for assessing trustworthiness of randomised controlled trials. *The BMJ*. <https://doi.org/10.1101/2025.09.03.25334905>

2025

- Anvari, F., Alsalti, T., Oehler, L., **Hussey, I.**, Elson, M., & Arslan, R.C. (2025). A fragmented field: Construct and measure proliferation in psychology. *Advances in Methods and Practices in Psychological Sciences*. <https://doi.org/10.1177/25152459251360642>
- Anvari, F., Alsalti, T., Oehler, L., **Hussey, I.**, Elson, M., & Arslan, R.C. (2025). Defragmenting psychology. *Nature Human Behavior*, 9, 836–839. <https://doi.org/10.1038/s41562-025-02138-0>
- Cummins, J., Elson, M., & **Hussey, I.** (2025). Cognitive dissonance in large language models is neither cognitive nor dissonant. *Proceedings of the National Academy of Sciences*, 122(35), e2517912122. <https://doi.org/10.1073/pnas.2517912122>
- Hussey, I.** (2025). Data is not available upon request. *Meta-Psychology*. <https://doi.org/10.15626/MP.2023.4008>
- Hussey, I.** (2025). Verification report: A critical reanalysis of Vahey et al. (2015) “A meta-analysis of criterion effects for the Implicit Relational Assessment Procedure (IRAP) in the clinical domain”. *Behavior Therapy and Experimental Psychiatry*. <https://doi.org/10.1016/j.jbtep.2024.102015>
- Hussey, I.**, Alsalti, T., Bosco, F., Elson, M., & Arslan, R. (2025). An aberrant abundance of Cronbach's alpha values at .70. *Advances in Methods and Practices in Psychological Science*, 8(1). <https://doi.org/10.1177/25152459241287123>
- Hussey, I.**, & Elson, M. (2025). Verification Report of Zhao et al. (2023) ‘Effect of Acceptance and Commitment Therapy for depressive disorders: a meta-analysis’. *Lifecycle Journal*. <https://www.researchhub.com/paper/9404372> https://doi.org/10.31234/osf.io/84e7t_v1

2024 and prior

- Röseler, L., Kaiser, L., Doetsch, C.A., Klett, K., Seida, C., ... **Hussey, I.**, ... & Zhang, Yikang. (2024) The Replication Database: Documenting the Replicability of Psychological Science. *Journal of Open Psychology Data*, 12(1). <https://doi.org/10.5334/jopd.101>

- Borgert, N., Reithmaier, O.D., Jansen, L., Hillemann, L., **Hussey, I.**, & Elson, M. (2023). Home Is Where the Smart Is: Development and Validation of the Cybersecurity Self-efficacy in Smart Homes (CySESH) Scale. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. <https://doi.org/10.1145/3544548.3580860>
- Corneille, O., Havemann, J., Henderson, E. L., IJzerman, H., **Hussey, I.**, Orban de Xivry, J., Jussim, L., Holmes, N. P., Pilacinski, A., Beffara, B., Carroll, H., E. L., Outa, N. O., Lush, P., J., & Lotter, L. D. (2023). Beware ‘persuasive communication devices’ when writing and reading scientific articles. *eLife*. <https://doi.org/10.7554/elife.88654>
- Elson, M., **Hussey, I.**, Alsalti, T., & Arslan, R. C. (2023). Psychological Measures Aren’t Toothbrushes. *Communications Psychology*. <https://doi.org/10.1038/s44271-023-00026-9>
- Hughes, S., Cummins, J., & **Hussey, I.** (2023). Effects on the Affect Misattribution Procedure are Strongly Moderated by Awareness. *Behavior Research Methods*, 1(29). <https://doi.org/10.3758/s13428-022-01879-4>
- Hussey, I.** (2023). A systematic review of Null Hypothesis Significance Testing, sample sizes and statistical power in research using the Implicit Relational Assessment Procedure. *Journal of Contextual Behavioral Science*, 29, 86-97. <https://doi.org/10.1016/j.jcbs.2023.06.008>
- Cummins, J., **Hussey, I.**, & Spruyt, A. (2022). The role of attitude features in the reliability of IAT scores. *Journal of Experimental Social Psychology*, 101, 104330. <https://doi.org/10.1016/j.jesp.2022.104330>
- Greenwald, A. G., Brendl, M., Cai, H., Cvencek, D., Dovidio, J. F., Friesse, M., ... & Wiers, R. W. (2022). Best research practices for using the Implicit Association Test. *Behavior Research Methods*, 1-20. <https://doi.org/10.3758/s13428-021-01624-3>
- Kurdi, B., **Hussey, I.**, Stahl, C., Hughes, S., Unkelbach, C., Ferguson, N., ... (2022). Unaware attitude formation in the surveillance task? Revisiting the findings of Moran et al.(2021). *International Review of Social Psychology*, 35(1). <https://doi.org/10.5334/irsp.546>
- Hussey, I.** (2021). A method to streamline *p*-hacking. *Meta Psychology*, 5. <https://doi.org/10.15626/MP.2020.2529>
- Kirtley, O. J., **Hussey, I.**, & Marzano, L. (2021). Exposure to and experience of self-harm: An exploratory network analysis. *Psychiatry Research*, 295, 113572. <https://doi.org/10.1016/j.psychres.2020.113572>
- Moran, T., Hughes, S., **Hussey, I.**, Vadillo, M. A. Olson, M. A., Aust, F., Bading, K., Balas, R., Benedict, T., Corneille, O., Douglas, S. B., Ferguson, M. J., Fritzlen, K. A., Gast, A., Gawronski, B., Heycke, T., Högden, F., Hütter, M., Kurdi, B., Mierop, A., Richter, J., Sarzyńska, J., Smith, C. T., Stahl, C., Thomasius, P., Unkelbach, C., & De Houwer, J. (2021). Incidental Attitude Formation via the Surveillance Task: A Pre-Registered Replication of Olson and Fazio (2001). *Psychological Science*, 32(1). <https://doi.org/10.1177/09567976209685>
- Cracco, E., González-García, C., **Hussey, I.**, Braem, S., & Wisniewski, D. (2020). Cultural pressure and biased responding in free will attitudes. *Royal Society Open Science*, 7(8), 191824. <https://doi.org/10.1098/rsos.191824>
- Hughes, S., De Houwer, J., Mattavelli, S., & **Hussey, I.** (2020). The Shared Features Principle: If Two Objects Share a Feature, People Assume Those Objects Also Share Other Features. *Journal of Experimental Psychology: General*, 149(12), 2264–2288. <https://doi.org/10.1037/xge0000777>
- Hughes, S., Mattavelli, S., **Hussey, I.**, & De Houwer, J. (2020). The influence of extinction and counterconditioning procedures on operant evaluative conditioning and intersecting regularity effects. *Royal Society Open Science*, 7(10), 192085. <https://doi.org/10.1098/rsos.192085>
- Hussey, I.** & Hughes, S. (2020). Hidden invalidity among 15 commonly used measures in Social and Personality psychology. *Advances in Methods and Practices in Psychological Science*, 3(2), 166-184. <https://doi.org/10.1177/2515245919882903>
- Landy, J., Jia, M., Ding, I. L., Viganola, D., Tierney, W., Dreber, A., ... **Hussey, I.**, ... & Uhlmann, E. L. (2020). Crowdsourcing hypothesis tests: Making transparent how design choices shape research results. *Psychological Bulletin*, 146(5), 451-479. <https://doi.org/10.1037/bul0000220>
- Van Dessel, P., De Houwer, J., Hughes, S., & **Hussey, I.** (2018). An Analysis of the Scientific

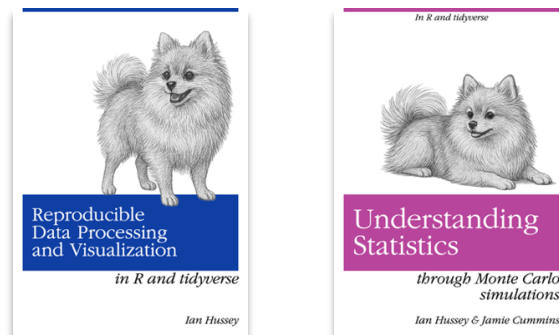
- Status and Limitations of the Attitudinal Entropy Framework and an Initial Test of Some of Its Empirical Predictions. *Psychological Inquiry*, 29(4), 213–217.
<http://doi.org/10.1080/1047840X.2018.1537339>
- Hussey, I.**, & De Houwer, J. (2018). The Implicit Association Test as an analogical learning task. *Experimental Psychology*, 65(5), 272–285. <https://doi.org/10.1027/1618-3169/a000416>
- De Schryver, M., **Hussey, I.**, Cartwright A., & Barnes-Holmes, D. (2018). The PI: an alternative scoring-algorithm for the IRAP using a more robust effect size measure. *Journal of Contextual Behavioral Science*, 7, 97–103. <https://doi.org/10.1016/j.jcbs.2018.01.001>
- Cartwright, A., **Hussey, I.**, Roche, B., Dunne, J., & Murphy, C. (2017). An investigation into the relationship between the gender binary and occupational discrimination using the Implicit Relational Assessment Procedure. *The Psychological Record*, 67(1), 121–130.
<http://doi.org/10.1007/s40732-016-0212-1>
- Finn, M., Barnes-Holmes, D., **Hussey, I.**, & Graddy, J. (2016). Exploring the behavioral dynamics of the Implicit Relational Assessment Procedure: The impact of three types of introductory rules. *The Psychological Record*, 66(2), 309–321.
<http://doi.org/10.1007%2Fs40732-016-0173-4>
- Hughes, S., **Hussey, I.**, Corrigan, B., Jolie, K., Murphy, C., & Barnes-Holmes, D. (2016). Faking revisited: Exerting strategic control over performance on the Implicit Relational Assessment Procedure. *European Journal of Social Psychology*, 46(5), 632–648.
<http://doi.org/10.1002/ejsp.2207>
- Hussey, I.**, Barnes-Holmes, D., & Booth, R. (2016). Individuals with current suicidal ideation demonstrate implicit fearlessness of death. *Journal of Behavior Therapy & Experimental Psychiatry*, 51, 1–9. <http://doi.org/10.1016/j.jbtep.2015.11.003>
- Hussey, I.**, Ní Mhaoileoin, D., Barnes-Holmes, D., Ohtsuki, T., Kishita, N., Hughes, S., & Murphy, C. (2016). The IRAP is nonrelative but not acontextual: Changes to the contrast category influence men's dehumanization of women. *The Psychological Record*, 66(2), 291–299. <http://doi.org/10.1007%2Fs40732-016-0171-6>
- Kavanagh, D., **Hussey, I.**, McEntegart, C., Barnes-Holmes, Y. & Barnes-Holmes, D. (2016). Using the IRAP to explore natural language statements. *Journal of Contextual Behavioral Science*, 5(4), 247–251. <http://doi.org/10.1016/j.jcbs.2016.10.001>
- Barnes-Holmes, D., & **Hussey, I.** (2015). The functional-cognitive meta-theoretical framework: Reflections, possible clarifications and how to move forward. *International Journal of Psychology*, 51(1), 50–57. <http://doi.org/10.1002/ijop.12166>
- Barnes-Holmes, D., Barnes-Holmes, Y., **Hussey, I.**, & Luciano, C. (2015). Relational Frame Theory: Finding its historical and intellectual roots and reflecting upon its future development. In R. D. Zettle, S. C. Hayes, D. Barnes-Holmes, & A. Biglan (Eds.), *The Wiley Handbook of Contextual Behavioral Science* (pp. 115–128). New York, NY: Blackwell-Wiley. <http://doi.org/10.1002/9781118489857>
- Barnes-Holmes, Y., **Hussey, I.**, McEntegart, C., Barnes-Holmes, D., & Foody, M. (2015). Scientific ambition: The relationship between Relational Frame Theory and middle-level terms in acceptance and commitment therapy. In R. D. Zettle, S. C. Hayes, D. Barnes-Holmes, & A. Biglan (Eds.), *The Wiley Handbook of Contextual Behavioral Science* (pp. 365–382). New York, NY: Blackwell-Wiley. <http://doi.org/10.1002/9781118489857>
- Hussey, I.**, Barnes-Holmes, D., & Barnes-Holmes, Y. (2015). From Relational Frame Theory to implicit attitudes and back again: clarifying the link between RFT and IRAP research. *Current Opinion in Psychology*, 2, 11–15. <http://doi.org/10.1016/j.copsyc.2014.12.009>
- Hussey, I.**, Daly, T., & Barnes-Holmes, D. (2015). Life is good, but death ain't bad either: Counter-intuitive implicit biases to death in a normative population. *The Psychological Record*, 65(4), 731–742. <http://doi.org/10.1007/s40732-015-0142-3>
- Hussey, I.**, Thompson, M., McEntegart, C., Barnes-Holmes, D., & Barnes-Holmes, Y. (2015). Interpreting and inverting with less cursing: A guide to interpreting IRAP data. *Journal of Contextual Behavioral Science*, 4(3), 157–162.
<http://doi.org/10.1016/j.jcbs.2015.05.001>
- McEntegart, C., Barnes-Holmes, Y., **Hussey, I.**, & Barnes-Holmes, D. (2015). Ties between basic science of language and cognition to clinical applications. *Current Opinion in Psychology*, 2, 56–59. <http://doi.org/10.1016/j.copsyc.2014.11.017>

- O'Regan, L. M., Farina, F. R., **Hussey, I.**, & Roche, R. A. (2015). Event-related brain potentials reveal correlates of the transformation of stimulus functions through derived relations in healthy humans. *Brain Research*, 1599, 168–177.
<http://doi.org/10.1016/j.brainres.2014.12.044>
- Hussey, I.**, & Barnes-Holmes, D. (2012). The Implicit Relational Assessment Procedure as a measure of implicit depression and the role of psychological flexibility. *Cognitive and Behavioral Practice*, 19(4), 573–582. <http://doi.org/10.1016/j.cbpra.2012.03.002>

Peer-reviewed publications related to the ERЯOR project I co-run

- Elson, M. (2024). A Bug Bounty Program for Science. *Nature*, 629, 730.
<https://doi.org/10.1038/d41586-024-01465-y>

Open Access textbooks



- Hussey, I.** (2026). *Reproducible data processing and visualization in R and tidyverse*. doi: [10.5281/zenodo.19427859](https://doi.org/10.5281/zenodo.19427859)
- Textbook for Master's seminar 'Reproducible data processing and visualization in *R* and tidyverse'
 - Book and code for exercises available at: <https://wrangling.tidyverse.se>
 - Includes web apps for interactive coding exercises – [link to example](#)
- Hussey, I.** & Cummins, J. (2026). *Understanding statistics through Monte Carlo simulations*. doi: [10.5281/zenodo.19420477](https://doi.org/10.5281/zenodo.19420477)
- Textbook for Master's seminar 'Improving your statistical inferences through Monte Carlo simulations in *R* and tidyverse'
 - Book and code for exercises available at: <https://simulations.tidyverse.se>

Research funding

Total grants awarded:	CHF 1.20 million
Total as PI:	CHF 1.07 million
Total as co-investigator:	CHF 128 thousand
2026	Project Funding (Primary Investigator) Swiss National Science Foundation <i>Title: Quantifying Effect Size Inflation in Clinical Psychology Trials (QUEST)</i> CHF 789'082
2026	Metascience Research Grant (International Co-Investigator) UK Research and Innovation <i>Title: 'After the Flood: Near-Future Scenarios and Countermeasures of AI-Accelerated Research Fabrication'</i> Co-investigators: - Prof. Sebastian Deterding, Imperial College London - Prof. Samuel Cooper, Imperial College London - Prof. Leila Sheldrick, Imperial College London CHF 109'000 (CHF 380'000 total award)
2026	Seed Grant: Research Synthesis and Meta-Research (Co-investigator) Center for Reproducible Science and Research Synthesis, University of Zurich <i>Title: ARRIVE-Check: Automated Assessment of ARRIVE Reporting Quality in Animal Studies Using NLP and LLM-Assisted Document Analysis</i> CHF 18'478
2024	Scientific Exchange (Primary Investigator) Swiss National Science Foundation Visiting Research Fellowship at the Oxford Internet Institute CHF 3'420
2024	Research funding for mid-level staff (Primary Investigator) University of Bern, Institute of Psychology CHF 952
2017	Postdoctoral scholarship (Primary Investigator) Ghent University Special Research Fund CHF 222'400
2012 – 2014	Government of Ireland Ph.D. scholarship (Primary Investigator/sole awardee) Irish Research Council CHF 52'200
2013	Summer Program for Undergraduate Research (Primary Investigator/sole awardee) Maynooth University Funding for a research assistant CHF 2'200

Supervision

Current research group

Postdocs and doctoral students

2026 – present Dr. Ze Freeman (postdoc)

- Expertise in clinical trials for psychological interventions
- Research on development and validation of statistical methods for quantifying effect size inflation in clinical trials (QUEST SNSF project)

2026 – present Wilfried Lakena (doctoral student)

- Mathematician with clinical trials and Trustworthiness Assessment expertise
- Second supervisor: Prof. Dr. Dr. Benjamin Ineichen, head of the Medical Data Science Unit, University of Bern.
- Research on development and validation of statistical methods for quantifying effect size inflation in clinical trials (QUEST SNSF project)

2025 – present Lukas Jung (doctoral student)

- Software engineer with Trustworthiness Assessment expertise
- Developer of the *scrutiny* and *unsum* R packages
 - lhdjung.github.io/scrutiny
 - lhdjung.github.io/unsum
- Research on development and validation of statistical methods for Trustworthiness Assessment

Master's students

4 students Monte Carlo simulation studies on the following topics:

- Trustworthiness of reported rounded correlation tables via positive-semi-definiteness
- Proportion of data tampering required to obtain significant results
- The influence of different forms of p -hacking on the Type I error rate
- The replicability of item-dropping recommendations based on Cronbach's-alpha-if-item-removed

Research group alumni

Doctoral students

2018 – 2021 Dr. Jamie Cummins (co-supervision at Ghent University)

PhD thesis title: 'Developing new measures of spontaneous beliefs'

Master's students

2024 – present 5 completed students at University of Bern

2018 – 2020 2 students at Ghent University

Bachelor's students

2023 – present 4 students at University of Bern

2022 – 2023 3 students at Ruhr University Bochum

2011 – 2015 23 students at Maynooth University

Research assistants

2023 – present 5 research assistants at University of Bern
 2022 – 2023 2 research assistants at Ruhr University Bochum
 2013 – 2014 2 research assistants at Maynooth University

Methods development

R packages

I am author and maintainer of the following *R* packages for **teaching coding in *R*** to groups of any size, and packages that **develop, validate, and implement statistical methods** for Trustworthiness Assessment.



Trustworthiness Assessments for reported summary statistics using *bounds* checks. Tests whether reported mean and Standard Deviation are consistent with constraints including the scale range, sample size, and integer data. Shiny web app [here](#).

[Link](#)



Trustworthiness Assessments for reported summary statistics using *recalculation* checks. Calculate bounds for a variety of tests' *p*-values, effect sizes, and parameter estimates from reported summary statistics given rounding. Shiny web apps [here](#) and [here](#).

[Link](#)



Trustworthiness Assessments for reported Mann-Whitney tests using *granularity* checks. Analytic solution for Grimes and Heather's (2026) GRIM-U test.

[Link](#)



Trustworthiness Assessments for reported correlation tables. Check whether reported correlation tables are Positive-Semi-Definite given rounding precision, and attempt to isolate the source of any issues to specific correlations. Shiny web app [here](#).

[Link](#)



Learn to wrangle messy data. Create datasets with known effects hidden in messy data for teaching data wrangling.

[Link](#)



Learn to keep data and *R* code organized and reproducible. Creates Quarto templates and file structures that are compliant with the Psych-DS data standard. Allows users to check and manage their project's compliance, with corrective feedback.

[Link](#)

iOS app development



turnstile

An iOS app and researcher platform to study digital behaviour through automated event-contingent and Automated Signal-Contingent EMA/ESM assessment and Just-In-Time Adaptive Interventions (JITAI).

turnstile-app.com

Statistical and methodological expertise

I regularly make use of, teach, and provide consultation for a wide range of statistical methods, from the core statistical skills of taught in the Bachelor's to advanced methods.

Statistics

- The general linear model family as taught across the psychology curriculum
 - o t -tests, ANOVA family, and regression as special cases
 - o Hypothesis testing and estimation approaches
 - o Model parameter based approaches and marginal/conditional prediction approaches
- Generalized linear models including logistic, ordinal, and beta regression
- Linear mixed effects/multilevel models
- Meta-analysis
- Latent variable and psychometric modelling
 - o Factor Analysis and structural validity assessment (Hussey & Hughes, 2020)
 - o Item Response Theory
 - o Network analysis (Kirtley et al., 2021)
- Bayesian modelling
 - o Visualisation and justification of regularising priors (*brms*/Stan; see Norwood et al., 2025)
 - o Regularised prediction and poststratification for test norming (Alsalti et al., 2026)
- Intensive longitudinal data
 - o Partialling of within and between effects; opportunities and threats to causal interpretation (Norwood et al., 2026)
- Causal inference from (non)experimental designs
 - o Mediation, moderation
 - o Modelling of confounds, colliders, and selection biases
 - o RCT methods: Intention To Treat, Average Treatment Effect, Average Treatment Effect on the Treated, etc.
- Simulation
 - o Individual simulated datasets for developing analyses prior to data collection (see my *truffle R* package)
 - o ADEMP-specified Monte Carlo studies (see Hussey & Cummins, 2026 textbook)
 - o Exhaustive enumeration / grid-search verification of deterministic methods
- Mathematical statistics
 - o Closed-form analytic derivation (see Hussey et al., 2026, the bounds of Standard Deviation under constraints)

Coding

- Reproducible code and literate programming (see *psychdsish R* package)
- Version control (git)
- Writing *R* packages
- High Performance Computing via UniBe's UBELIX computing cluster
- LLM usage via API access
- Research tool development (Shiny apps, LLM and agentic tools)

Teaching

Bachelor's and Master's teaching

At UniBe

I have been responsible for designing the research methods and statistics curriculum for the Master's focus 'Psychology of Digitalisation'.

Title	Years	Type
Psychology of Digitalisation (link to syllabus)	2025 –present	Master's lecture ¹
Digitalisation of Psychology	2025 –present	Master's lecture ¹
Improving your statistical inferences through Monte Carlo simulations in <i>R</i> and tidyverse	2024 –present	Master's methods seminar ^{*,2}
Reproducible data processing and visualization in <i>R</i> and tidyverse	2023 –present	Master's methods seminar
Estimating the credibility of past research	2023 –present	Master's methods seminar [*]
Psychology in crisis? Replications and reform movements in psychological science	2023 –present	Master's methods seminar ^{*,3}
The science of behavior: An introduction to behaviorism in principle and practice	2023 –2024	Bachelor's & Master's seminars ^{*,4}

Notes:

* Awarded a faculty teaching award for teaching evaluations in the top 4%

¹ co-delivered with Prof. Malte Elson and Dr. Sandra Grinchgl

² co-delivered with Dr. Jamie Cummins

³ co-delivered with Prof. Malte Elson

⁴ co-delivered with Dr. Sabrina Norwood

Courses taught at other institutions

2023	Psychology in crisis? An introduction to the Open Science movement and the evidence it has generated Bachelor's level seminar Ruhr University Bochum
2023	Research orientated internship Bachelor's level seminar Ruhr University Bochum
2011	Research methodology and statistics using SPSS Master's level seminar Trinity College Dublin
2010 – 2014	Visual Basic programming for experimental psychology Bachelor's level seminar Maynooth University

Doctoral level teaching

2025	How to conduct systematic reviews of research methods aka “meta-methods reviews” 1-week doctoral level workshop 9 th Trifels Summer School on Open Science, University of Mannheim
2024	Simulated data and Monte-Carlo simulation studies 2-day doctoral level workshop Doctoral Program Brain and Behavioral Sciences University of Bern
2023	How to conduct systematic reviews of research methods aka “meta-methods reviews” 1-week doctoral level workshop 8 th Trifels Summer School on Open Science, University of Mannheim
2017	Using PsychoPy for experimental psychology research 1-day doctoral level workshop Ghent University
2013 – 2014	Using PsychoPy for experimental psychology research 1-day doctoral level workshop Maynooth University

Research awards

2026	Commendation from the Society for the Improvement of Psychological Science <i>Awarded for:</i> RegCheck (https://regcheck.app) <i>Awarded to:</i> Jamie Cummins, Ian Hussey, Beth Clarke, and Malte Elson
2025	Commendation from the Society for the Improvement of Psychological Science <i>Awarded for:</i> ERJIOR: a Bug-Bounty Program for Science <i>Awarded to:</i> Ian Hussey, Malte Elson, Ruben Arslan, and Brian Nosek, Sean Hughes, Jordan Axt, Calvin Lai, and Charlie Ebersole
2024	Pubpeer comment award <i>Awarded for:</i> Comments providing significant new scientific insight into a publication (link) <i>Awarded to:</i> The ‘Cake Club’ journal club of Psychology of Digitalisation, University of Bern
2020	Commendation from the Society for the Improvement of Psychological Science <i>Awarded for:</i> “the curation of the Attitudes, Identity, and Individual Differences dataset: a large dataset with high reuse potential within the behavioral sciences”, which has since been used in a large number of Registered Reports. <i>Awarded to:</i> Ian Hussey, Brian Nosek, Sean Hughes, Jordan Axt, Calvin Lai, and Charlie Ebersole.
2013	‘Making an Impact’ award from the Irish Higher Education Authority <i>Awarded for:</i> “research likely to make a difference to a particular aspect of Irish life, to the country as a whole, or internationally” <i>Awarded to:</i> Ian Hussey

Keynote speaker and invited talks

- 2026 (planned) The emerging field of Forensic Meta-Science
The Association for Interdisciplinary Meta-Research and Open Science (AIMOS)
Victoria University of Wellington
- 2026 (planned) Trustworthiness Assessment: Cochrane's new procedure for excluding
 untrustworthy studies from research syntheses
Faculty of Medicine and Health
University of Sydney
- 2026 (planned) Taboo topics we can no longer avoid: negligence, recklessness, and fraud
Ringvorlesung: Fake Science - Isolated Cases or a Systemic Problem?
University of Zurich
- 2026 Making Trustworthiness assessment tools trustworthy
Workshop on Papermills and Scientific Misconduct
QUEST Center for Responsible Research
Berlin Institute of Health
- 2025 Taboo topics we can no longer avoid: negligence, recklessness, and fraud
Reproducible Research Cambridge
University of Cambridge
- 2025 Normalising error detection and Trustworthiness Assessment in science
Open Research Summer School
Kings College London
- 2025 Taboo topics we can no longer avoid: negligence, recklessness, and fraud
ReproducibiliTea Basel
University of Basel
- 2024 ERJØR: A three-pronged effort to improve error detection in science
Digital Mental Health Group
University of Cambridge
- 2024 ERJØR: Making scientific error detection easy, normative, and rewarding
Boost and Exchange Academic Conference (BEACON)
University of Basel
- 2024 ERJØR: A bug bounty program for science
Open Research Summer School
Kings College London
- 2024 The ERJØR project: A three-prong effort to improve post publication critique
 and error detection
Perspectives on Scientific Error meeting
TU Eindhoven
- 2023 Measurement Schmeasurement: How I learned to appreciate measurement
University of Bern
- 2023 (Sch)measurement in implicit measures
Catholic University of Louvain
Co-author: Jamie Cummins

- 2022 The best theory is a flawed one: lessons from implicit bias research
RIOT Science Club
- 2020 The best theory is a flawed one: Looking back on 25 years of research on implicit attitudes
University of Lincoln
- 2018 Open Science for experimental-clinical researchers
Centre for Contextual Psychiatry
KU Leuven
- 2018 Never waste a good (replication) crisis: Open Science for early career researchers
Early and Mid-Career Researchers' (EMCR) forum for suicide researchers,
University of Glasgow
- 2017 The Implicit Association Test as an analogical learning task
KU Leuven
- 2016 Using meta-theory to foster cooperation between behavioral and cognitive approaches to psychology
University of Chester

Expert meetings

The following is a list of by-invitation expert group meetings I have attended

- 2026 (planned) AI and Research Integrity Meeting
Organised by Prof. Brian Nosek
Centre for Open Science (COS)
Montreal
- 2026 (planned) Scaling Up Responsibly: The Ethics Of Forensic Peer Review
Organised by Prof. Irene Pasquetto
University of Maryland / Lake Como School of Advanced Studies
- 2024 Psych Garage: Initiatives to improve research practices in psychological science
Organised by Prof. Simine Vazire
University of Melbourne
- 2017 Function-Cognitive Framework Meeting
Ghent University

Conferences co-organised

- 2027 (planned) 2nd Fostering Accountability for the Integrity of Research Studies (FAIRS) meeting
sjcfairsmeeting.com
University of Bern
- 2025 7th Perspectives on Scientific Error meeting
errorsin.science
University of Bern

Science communication, media coverage, and policy citations

Policy citations

- 2026 *“The State of Scientific Publishing: Assessing Trends, Emerging Issues, and Policy Considerations”*
 Testimony given to the US House of Representatives
 Committee on Science, Space and Technology
 Investigations and Oversight Subcommittee Hearing
 By Kate Travis, Managing Editor of Retraction Watch
science.house.gov/2026/4/investigations-and-oversight-subcommittee-hearing-paper-mills
[Direct link](#) to the written testimony citing our work on INPECT-SR

Media coverage

- 2024 *Nature*
 ‘Cash for errors: project offers bounty for spotting mistakes in published papers’
 By Julian Nowogrodzki
<https://doi.org/10.1038/d41586-024-02681-2>
- 2024 *Science*
 ‘Wanted: scientific bounty hunters’
 News in Brief
www.science.org/content/article/news-glance-moon-landing-scientific-bounty-hunters-and-postdocs-facing-hunger
- 2024 *Science*
 ‘Cash For Published Errors’
 In the Pipeline
 By Derek Lowe
<https://www.science.org/content/blog-post/cash-published-errors>
- 2024 *The Chronicle of Higher Education*
 ‘Wanted: Scientific Errors. Cash Reward’
 By Stephanie M. Lee
www.chronicle.com/article/wanted-scientific-errors-cash-reward
- 2024 *WIRED Magazine*
 ‘Science Is Full of Errors. Bounty Hunters Are Here to Find Them’
 By Matt Reynolds
<https://www.wired.com/story/bounty-hunters-are-here-to-save-academia-bug-bounty/>
- 2024 *Süddeutsche Zeitung*
 ‘Das soll kein Pranger werden’ [‘It shouldn’t be a pillory’]
 By Sebastian Herrmann
<https://www.sueddeutsche.de/wissen/fehlerkultur-psychologie-1.6385692>
- 2024 *The Transmitter*
 ‘Bounty hunting for blunders’
 By Calli McMurray
<https://www.thetransmitter.org/publishing/bounty-hunting-for-blunders-qa-with-russell-poldrack-and-jan-wessel/>

2024 *Horizons*
 ‘Mistakes for sale’
 By Florian Fisch
<https://www.horizons-mag.ch/2024/06/06/mistakes-for-sale/>

Podcasts and radio

2024 *The Body of Evidence*
 ‘Bounty Hunting for Science’
<https://bodyofevidence.ca/bounty-hunting-for-science-with-ian-hussey>

2024 *Répare Ta Science*
 ‘The data is ultimately not available’
<https://open.spotify.com/episode/3xY7kp0RtK3a3pp2wE3a2H>

2015 RTE Radio 1 – *Bright Sparks*

2013 RTE Radio 1 – *Drivetime*

2013 Today FM Ireland – *News*

2013 SPIN 103.8 Ireland – *News*

2013 Kildare FM – *News*

Media training

2013 Direct instruction in media training and press release writing
Maynooth University Communications Office

Editorial work

Editorial boards

2024–present Chief Recommender
 ERJOR: A bug bounty program for science
error.reviews
 See Elson, M. (2024). A Bug Bounty Program for Science. *Nature*,
[10.1038/d41586-024-01465-y](https://doi.org/10.1038/d41586-024-01465-y)

2017–2020 Editorial Board
Journal of Contextual Behavioral Science

Editorial work

2025 Moderator
PsyArXiv

2012–2013 Editor
 Student Research Award
Association for Psychological Science Student Caucus

2010–2012 Associate Editor
Journal of European Psychology Students